Automated Flood Mapping Using Sentinel-2A Imagery and Machine Learning Classification Models

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Advanced Machine Learning Techniques

- Introduction to Machine Learning.
- Unsupervised learning.
- Regression and classification.
- Artificial Neural Networks.
- Advanced ML Techniques (CNNs, BDTs, etc.)
- Hands-on practice with Pytorch
- Applied ML in case studies
- Ethics in AI and ML
- Projects

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Background

- Between March and May 2024, severe flooding in Kenya resulted in at 291 deaths, 188 injuries, and 75 missing persons (NDOC).
- Heavy rains displaced 278,380 people (55,676 families) and affected 412,763 people (82,552 families) nationwide.
- The floods caused loss of 11,311 livestock, damage to 47,578 acres of croplands, 67 roads, 1,023 small businesses, and 129 schools (KRCS).
- At least 178 displacement sites were active, accommodating 71,704 people as of mid-May 2024.
- Areas include Narok, West Pokot, Nyeri, Siaya, Kirinyaga and, Nairobi.
- In Nairobi and surrounding areas, flooding disrupted socio-economic activities and inflicted significant human and material tolls.

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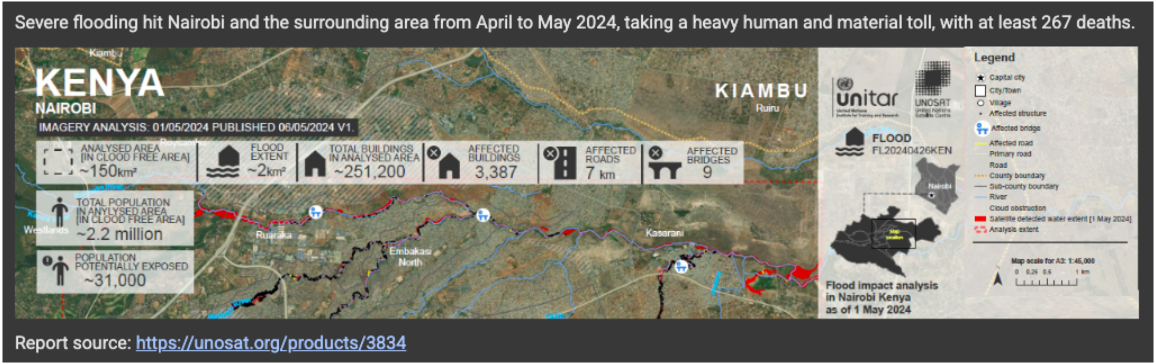
Severe flooding hit Nairobi and the surrounding area from April to May 2024, taking a heavy human and material toll, with at least 267 deaths.
KENYA
NAIROBI
IMAGERY ANALYSIS: 01/05/2024 PUBLISHED 06/05/2024 V1.
ANALYSED AREA [IN CLOOD FREE AREA] ~150km²
FLOOD EXTENT ~2km²
TOTAL BUILDINGS IN ANALYSED AREA ~251,200
AFFECTED BUILDINGS 3,387
AFFECTED ROADS 7 km
AFFECTED BRIDGES 9
TOTAL POPULATION IN ANYLYSED AREA [IN CLOOD FREE AREA] ~2.2 million
POPULATION POTENTIALLY EXPOSED ~31,000
KIAMBU
Ruiru
Ruaraka
Embakasi North
Kasarani
Westlands
unitar
UNOSAT
FLOOD FL20240426KEN
Nairobi
Flood impact analysis in Nairobi Kenya as of 1 May 2024
Legend
Capital city
City/Town
Village
Affected structure
Affected bridge
Affected road
Primary road
Road
County boundary
Sub-county boundary
River
Cloud obstruction
Satellite detected water extent [1 May 2024]
Analysis extent
Map scale for A3: 1:45,000
Report source: https://unosat.org/products/3834

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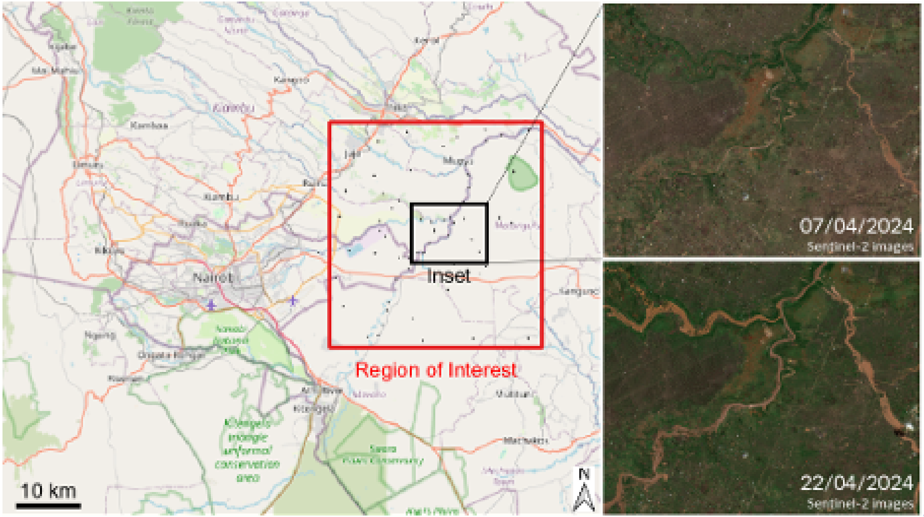
Problem Statement

- Flood assessment in Nairobi often relies on slow, manual methods with limited spatial detail.
- Sentinel-2 imagery provides detailed observations but is not directly usable without complex pre-processing.
- There is a need for automated machine learning approaches to transform satellite data into accurate flood maps for rapid decision-making.

Key Objective

- To develop an automated flood mapping framework using Sentinel-2 imagery and supervised machine learning.
- To generate accurate and timely flood extent maps for Nairobi to support disaster response and planning.

Study Area


Region of Interest
Inset
Nairobi
10 km
N
07/04/2024
Sentinel-2 images
22/04/2024
Sentinel-2 images

Methodology Flow: Data Preparation

Sentinel-2 Imagery

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Resampling

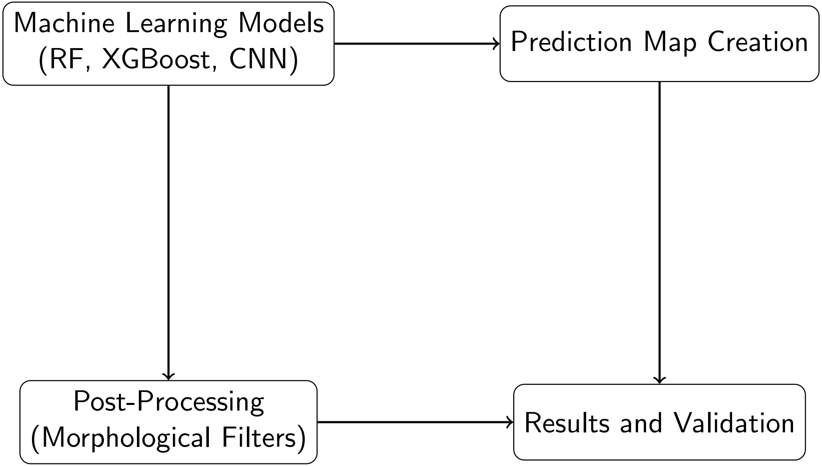
↓

Feature Engineering
(NDVI, NDWI, etc.)

↓

Normalization

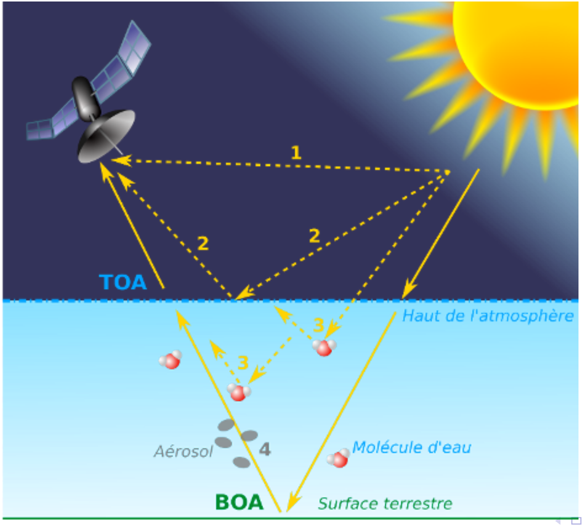
Methodology Flow: Modeling and Results



Data Download (Copernicus Open Access Hub)

- Copernicus browser provides Sentinel-2 images as 110 $\times$ 100 km tiles.
- Images are available at different pre-processing levels:
- **Level-1C**: Top of Atmosphere (TOA) reflectance, orthorectified, in UTM units.
- **Level-2A**: Bottom of Atmosphere (BOA) reflectance (after atmospheric correction), orthorectified, in UTM units.
- Sentinel-2 is a two-satellite constellation (Sentinel-2A launched in 2015, Sentinel-2B in 2017) operated by ESA under the Copernicus program.
- Provides free, open-source, high-resolution land surface imagery (10–60 m).
- Offers high temporal revisit: 10 days with one satellite, 5 days with both at mid-latitudes.
- Captures radiation from visible to near- and mid-infrared ranges.
- For this study, we utilize the BOA

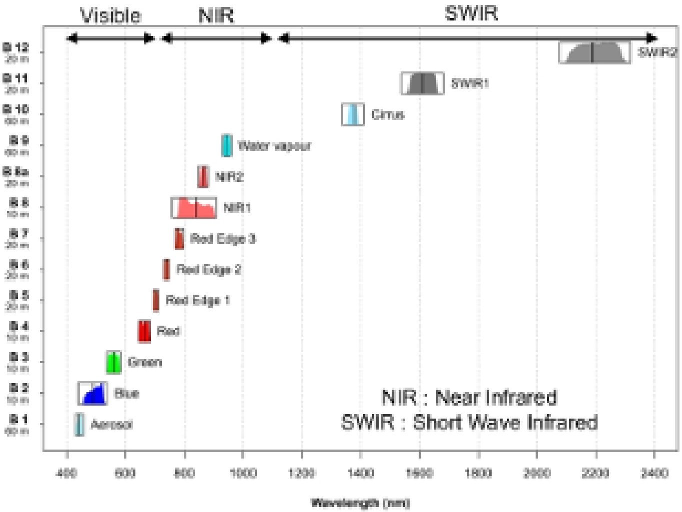
Data Download (Copernicus Open Access Hub)



Sentinel Bands

- A band is a specific range of wavelengths of the electromagnetic spectrum that a satellite sensor measures.
- Bands capture different information about the Earth's surface because materials (water, soil, vegetation, built-up areas) reflect and absorb radiation differently at different wavelengths.
- By combining or analyzing bands, we can detect features such as vegetation health, water bodies, soil moisture, floods, and urban areas.
- Delivers data in 13 spectral bands, with spatial resolutions of 10–60 m.

Sentinel Bands



Resampling

- The first step in pre-processing is downloading Sentinel-2 images.
- Bands originally at different resolution (20 to 60m) are resampled to 10 m.
- To ensure all bands have the same spatial resolution (10 m) for consistent analysis.
- After resampling, images are stored as a 3D stack: (number of bands, image height, image width).
- To align all spectral bands into a single data cube, making pixel-by-pixel multi-band analysis accurate and efficient.

Feature Engineering

- Feature engineering focuses on extracting relevant features for flood detection using spectral indices.
- Classical indices include:
  - **NDVI** (Normalized Difference Vegetation Index): distinguishes vegetation from non-vegetation.
  - **NDWI** (Normalized Difference Water Index): enhances water body detection.
- Additional indices for flood mapping:
  - **MNDWI** (Modified NDWI): improves water–built-up separation.
  - **AWEI** (Automated Water Extraction Index): minimizes shadow effects in water detection.
  - **NDFI** (Normalized Difference Flood Index): specifically designed to highlight flood-inundated areas.
- Indices are computed from Sentinel-2 spectral bands (Green, NIR, SWIR, etc.).

Feature Engineering

Bands Indices for flood mapping (Gao, 1996)

$$MNDWI = \frac{B03 - B011}{B03 + B011}$$

$$NDVI = \frac{B08 - B04}{B08 + B04}$$

$$NDFI = \frac{B04 - B012}{B04 + B012}$$

Normalization

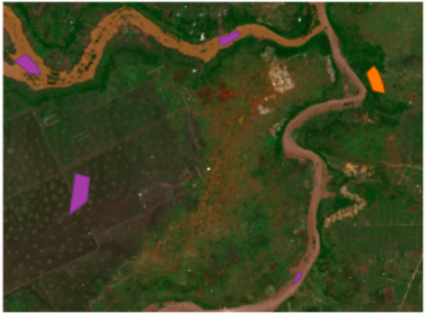
- After pre-processing, satellite images are transformed into a training dataset for machine learning models.
- Data normalization is required to ensure comparability across spectral bands.
- Normalization is performed band by band, standardizing values to a consistent scale.
- This step guarantees that training, testing, and prediction datasets share the same statistical distribution.
- Consistent normalization improves model stability, classification accuracy, and reliability of prediction maps.
- According to Alcaras (2025) normalization + index calculation enhances the detection of flooded and sediment-laden areas by reducing spectral bias.

Data Construction

- Training and testing zones are defined with associated labels (flooded vs. non-flooded).
- Polygons were created in QGIS.
- Pixels within polygons are selected, and reflectance values for each band are retrieved.
- A tabular dataset is generated: rows = samples, columns = features (band values).
- Pixels from flooded areas (label = 1) and non-flooded areas (label = 0) are extracted and grouped into training and texting.
- **Purpose:** Transform satellite imagery into pixel-based datasets for supervised machine learning, enabling accurate flood detection and mapping.

Data Construction

Create polygons with labels



Keep pixels inside polygons

Samples

B02	B03	B04	B04	B05

Model Specification

Supervised Machine Learning Models

- Decision Tree for XGBOOST
- Random Forest Bagging
- ADA Boost for decision Tree
- KNN.
- Linear Support Vector ML
- CNN 1D

Model Evaluation

- Best Model was selected based on the accuracy

Prediction Map Creation

Initial Map

- A trained machine learning model is applied to the normalized image stack.
- This generates a pixel-wise classification of flooded and non-flooded areas.
- An initial overview of the predictions provides spatial distribution of flood extent.

Post-Processing

- Morphological operators enhance the quality of classification results.
- **Opening (erosion followed by dilation):** removes small objects and separates connected regions.
- **Closing (dilation followed by erosion):** fills gaps and reconstructs objects with small voids.
- These refinements improve the accuracy and spatial coherence of flood maps.

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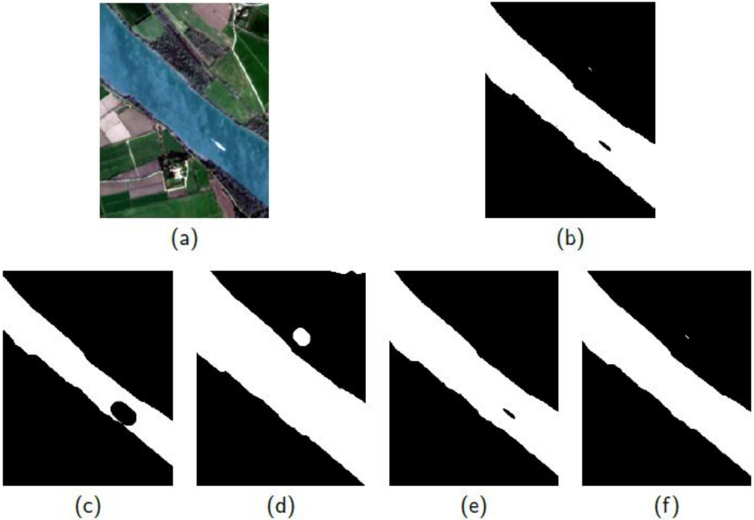
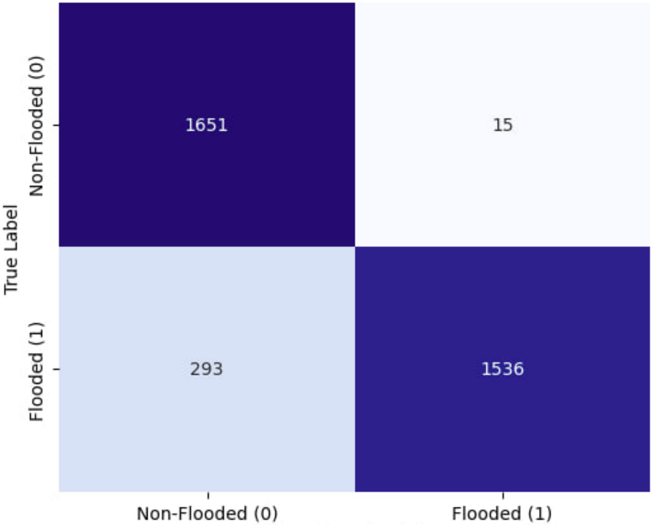


Model Selection

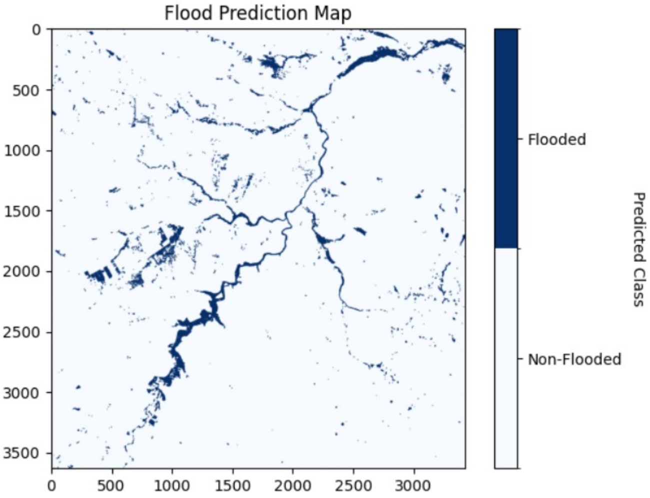
Table: Classification Accuracy of Different Machine Learning Models

Model	**Accuracy (%)**
K-Nearest Neighbors (KNN)	84.0
Support Vector Machine (LSVM)	86.1
CNN 1D	89.6
XGBOOST for decision Tree	90.6
Random Forest	90.9
Gradient BOOSTING	91.1
Random Forest with Bagging	**91.6**
AdaBoost with Decision Tree	91.0

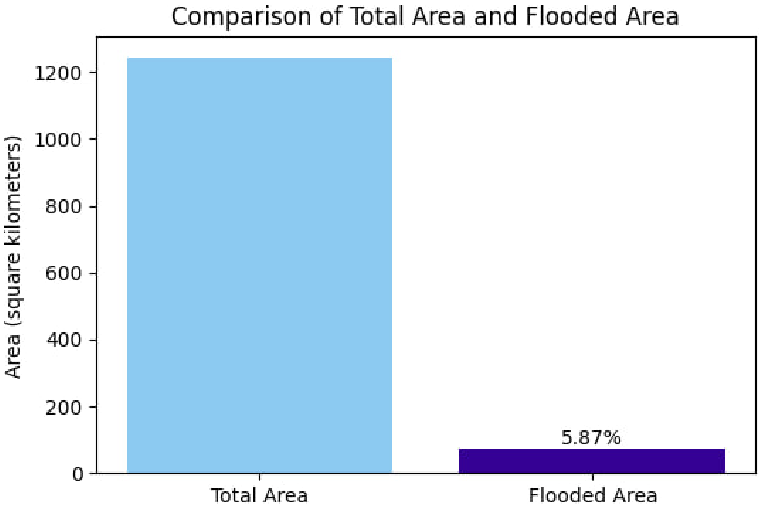
Confusion Matrix for rf



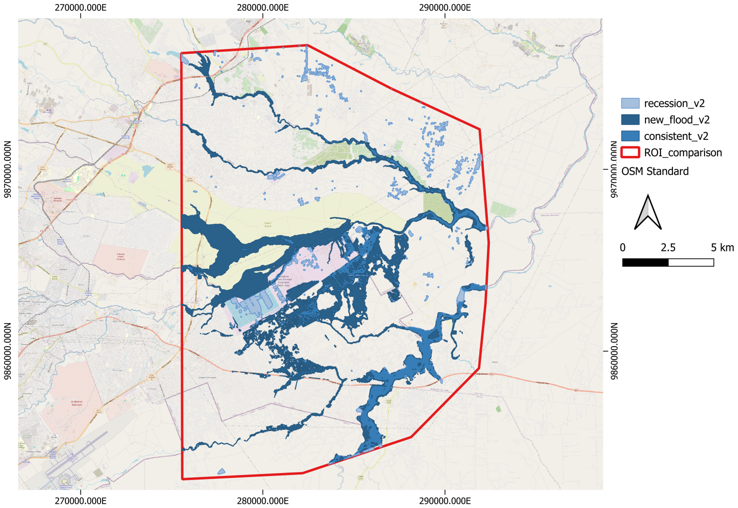
Prediction Map



Estimated Flooded Area



Comparison with Government Records



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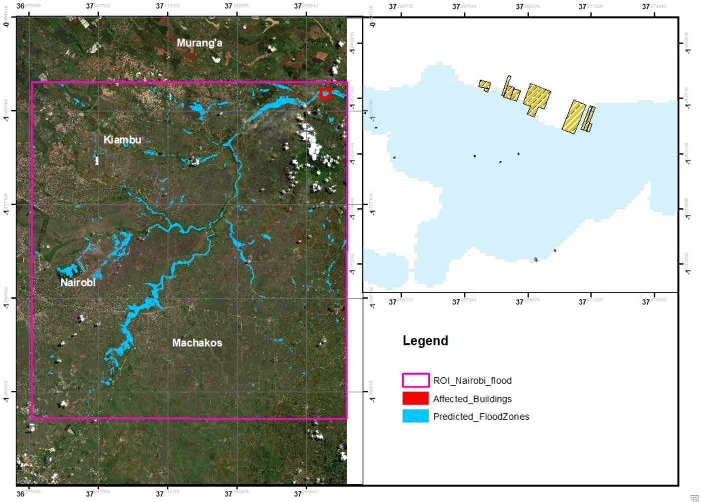
ROI Comparison

- New-flooded: areas where we predicted 0 (not flooded) but the government predicted 1(flooded)
- Consistent: areas where both predicted 1(flooded)
- Recession: areas where our study predicted 1(flooded) but government predicted 0(not flooded)

Total Number of Building in 110



Conclusion and Future Direction

Conclusion

- Sentinel-2 imagery integrated with machine learning enables accurate and automated flood mapping.
- Among the tested models, Random Forest with Bagging achieved the highest accuracy (91.6%).
- The predicted flood extent map identified widespread inundation consistent with government records.
- Analysis revealed significant socio-economic impacts, including damage to over 110 buildings in the study area.

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Future Direction

- Extend the analysis period to cover flood events from 2015 to 2025 for long-term trend evaluation.
- Incorporate more advanced machine learning models for improved classification.
- Expand the study to the entire Nairobi City to assess broader socio-economic impacts.
- Integrate watershed and drainage system analysis to understand hydrological drivers of flooding.
- Develop predictive tools to anticipate future flood events and support proactive urban planning and disaster preparedness.

Thank You

Asante Sana

I appreciate your time and attention.